

1) Given: F is a field of complex no.s
 V is a subspace of
polynomials over F .

To prove:

setⁿ: $V = \{ a_0 + a_1x + a_2x^2 + \dots + a_nx^n \mid a_i \in F, n \geq 0 \}$

~~Let B be~~

$$F \subseteq \mathbb{C}.$$

Let Basis of $V = X$,

$$X = \{ 1, x, x^2, x^3, \dots \}.$$

To prove:

- i). X is linearly independent
- ii). X spans V .

Proof:

i). Let linear combination of X :

$$c_0(1) + c_1(x) + c_2(x^2) + \dots + c_n x^n$$

$n \geq 0$

$$c_i \in \mathbb{R}, i \in [0, n]$$

Now,

$$\sum_{i=0}^n c_i x^i = 0$$

Inductive proof:

Base case: $\begin{cases} c_1(1) = 0 \\ c_1 = 0 \end{cases}$

Inductive case hypothesis:

for any k ,

$$c_1(1) + c_2 x^2 + \dots + c_k x^k = 0,$$

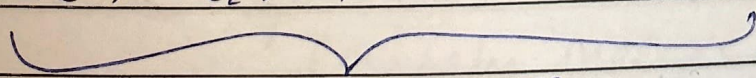
(1)

$$c_1, c_2, \dots, c_k = 0$$

Inductive case:

for $k+1$,

$$c_1(1) + c_2 x^2 + \dots + c_k x^k + c_{k+1} x^{k+1} = 0$$



$= 0$ from (1)

$$= 0 + c_{k+1} x^{k+1} = 0$$

$$\therefore \boxed{c_{k+1} = 0}$$

$$\therefore c_0 = c_1 = \dots = c_n = 0$$

$\therefore$ X is linearly independent.

(ii). X spans V .

every polynomial in V is

$$f(n) = a_0(1) + a_1(n) + a_2(n^2) + \dots + a_n(n^n)$$

~~However,~~

Linear combination of $X =$

$$c_0(1) + c_1(n) + c_2(n^2) + \dots + c_n(n^n)$$

substituting ~~$a_i = c_i$~~ , $c_i = a_i$

Linear combination of $X =$

$$a_0(1) + a_1(n) + a_2(n^2) + \dots + a_n(n^n)$$

$\therefore$ ~~V is a finite~~
for $v \in V$,

v is a finite linear combination

∞ elements in X .

$$\therefore \boxed{\text{span}(X) = V.}$$

Hence proved, $X = \{1, n, n^2, \dots\}$ is the basis of V , where ~~$\mathbb{R}$~~

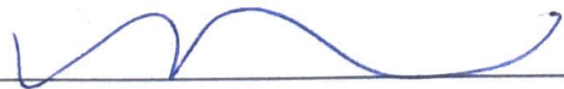
$$V = \left\{ a_0 + a_1 n + \dots + a_n n^n \right\} \left. \begin{array}{l} a_i \in F, \\ n \geq 0 \end{array} \right\}$$

$$F \subseteq \mathbb{C}.$$

2). Given: V is a vector space spanned by $\beta_1, \dots, \beta_m$.

To prove: An independent set of Vectors in V ~~MUST~~ be finite and cannot contain more than m elements.

Proof: Let $X = \{ \alpha_1, \dots, \alpha_n \}$



linearly independent.

Let $X = \{ \beta_1, \dots, \beta_m \}$
 spans V .

To prove: $m \geq n$.

proof:

$$\text{In } AX = 0$$

A is $m \times p$ and X is $p \times 1$.

$$\cancel{p \geq m},$$

~~then~~ $A = \{ a_{11}, a_{12}, \dots \}$

~~Proof~~

coefficients (scalars)

Q We can prove, if $p > m$, then

$$\{ v_1, \dots, v_p \}$$

linearly dependent

i.e. for $\sum_{i=1}^p c_i v_i$, $\exists c_j$ where $c_j \neq 0$.

$$c_j \in X.$$

γ is a spanning set. $\therefore$ Any vector ' u ' can be written as a linear combination of spanning set (γ).
 $(u \in V)$

$$\therefore u_j = \sum_{i=1}^n A_{ij} \beta_i, \quad j \in [1, p].$$

$$\sum_{j=1}^p c_j u_j = \sum_{j=1}^p c_j \sum_{i=1}^n A_{ij} \beta_i$$

$$\sum_{j=1}^p c_j u_j = \sum_{i=1}^n \left(\underbrace{\sum_{j=1}^p c_j A_{ij}}_{\text{scalar}} \right) \underbrace{\beta_i}_{\text{vector}}$$

But, if $p > n$, $AX=0$ has a

NON-TRIVIAL solⁿ & Theorem 6.4,

$$\sum A_{ij} x_j = 0 \text{ for } x \neq 0.$$

$$= \sum_{i=1}^n (0) (\beta_i) = 0$$

$$\therefore \sum c_j u_j = 0 \text{ for some } c_j \neq 0$$

$\therefore$ it is NOT linearly independent.

∴ Proof by contradiction!

p cannot be $> n$, else
linear independence fails.

substituting ~~$n \rightarrow p$~~ $p \rightarrow n$,

Hence proved, $\boxed{m \geq n.}$

3) Given: W and V are finite dimensional vector spaces such that $W \subsetneq V$.

To prove: $\dim W < \dim V$.

Proof: Let β be a basis of W .

~~Let~~ $\beta = \{w_1, w_2, \dots, w_k\}$

where $\text{span}(\beta) = W$ and β is linearly independent.

now, as $W \subsetneq V$,

there's at least one vector v_i ,
such that .

$$v_i \in V, v_i \notin W.$$

$$\therefore v_i \in V \setminus W.$$

Now, let $\text{Basis of } V =$

$$\beta_2 = \{v_1, v_2, \dots, v_k, v_{k+1}, \dots, v_n\}.$$

But, if W is a subspace of a
finite dimensional vector space V ,
every linearly independent subset of
 W is finite and part of a finite
subset of W .

2).

Theorem 5 - Hoffman and Kunze.

$$\therefore \beta_1 \subset \beta_2$$

$$\therefore \beta_2 = \{w_1, w_2, \dots, w_k, v_{k+1}, \dots, v_n\}$$

$$\therefore n > k.$$

Now, dimension of $V = \dim V = n$
dimension of $W = \dim W = k$

$\therefore$ dimension of $X =$ no. of elements in
basis of X .

$$\therefore n > k$$

$$\therefore \boxed{\dim V > \dim W.}$$

Hence proved.